SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	**Assessment:**	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO5 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)*		
Student Name:	Y.Krishna Tejesh Reddy	**Reg. No.:**	192411200
Section / Batch:	Slot C	**Date:**	29-7-2026

Code of Conduct

I _Y . Krishna Tejesh Reddy(192411200)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __ ________________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
	GRAND TOTAL:	**20 Marks**	

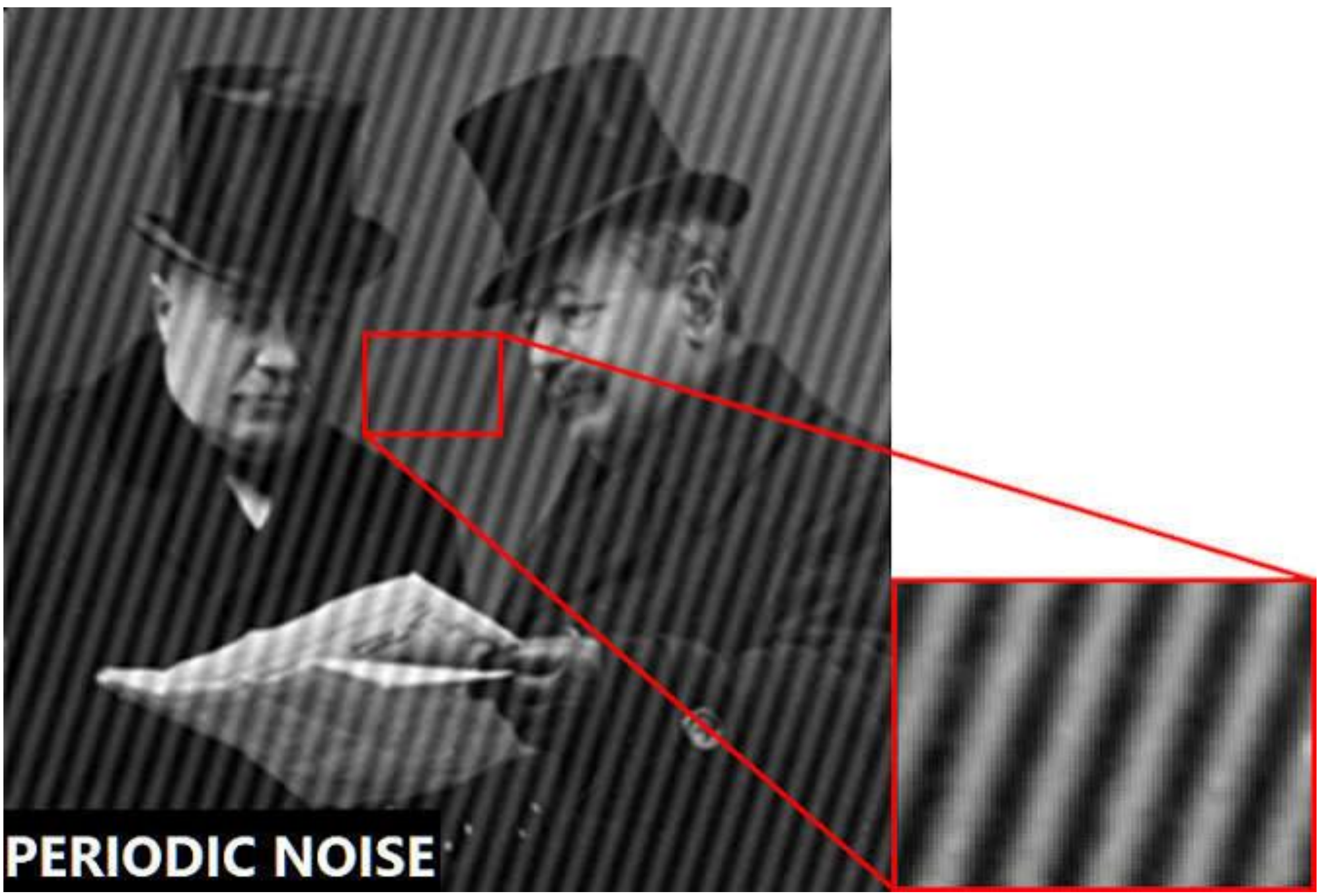
Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

a) Interpret the scenario and identify the type of noise present.

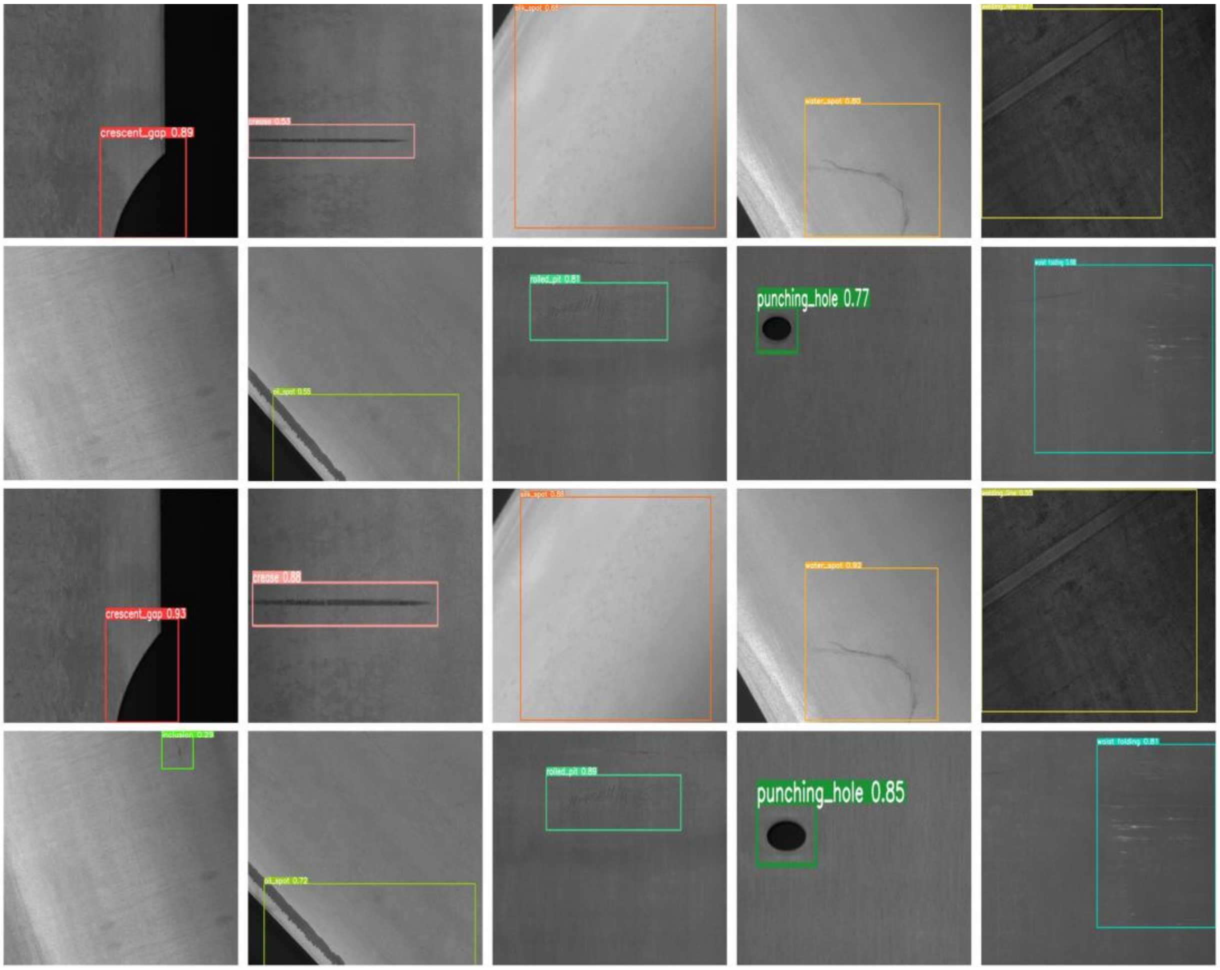
An automated quality inspection system captures images of metal components. Due to electrical interference, repetitive horizontal stripes appear across the images. This is known as **Periodic Noise**, which occurs because of electrical or electromagnetic interference during image acquisition. In the frequency domain, this noise appears as distinct peaks.



b) Explain how this degradation affects inspection accuracy.

Periodic noise hides cracks, scratches, and surface defects present on the metal component. It lowers image quality, making defect detection algorithms unreliable. As a result, defective products may be accepted, while good products may be rejected, reducing manufacturing efficiency.

YOLOv5s



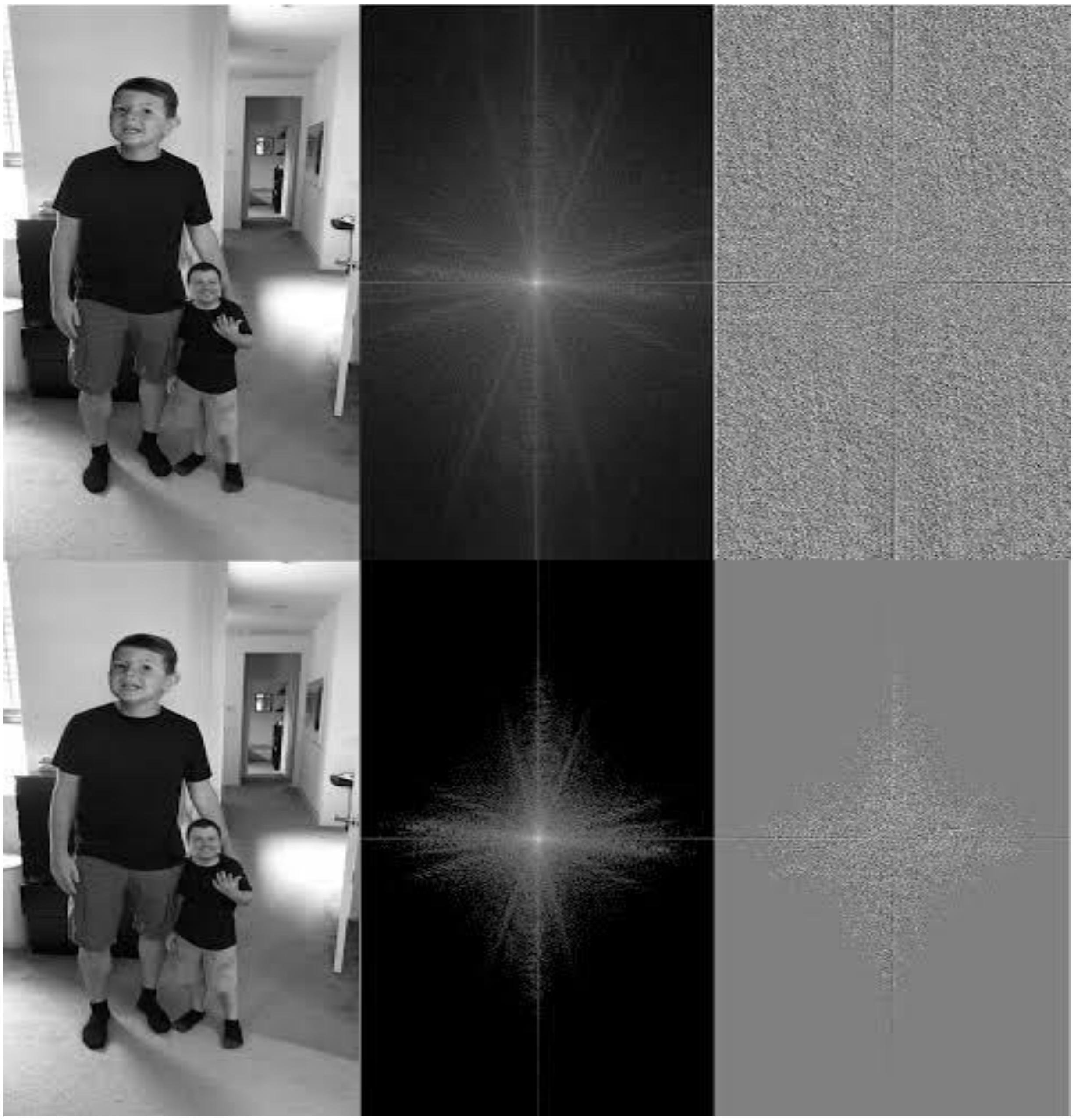
Ours

c) Recommend an appropriate enhancement technique to remove the interference.

The most suitable enhancement technique is the **Notch Filter** applied in the **Frequency Domain**.

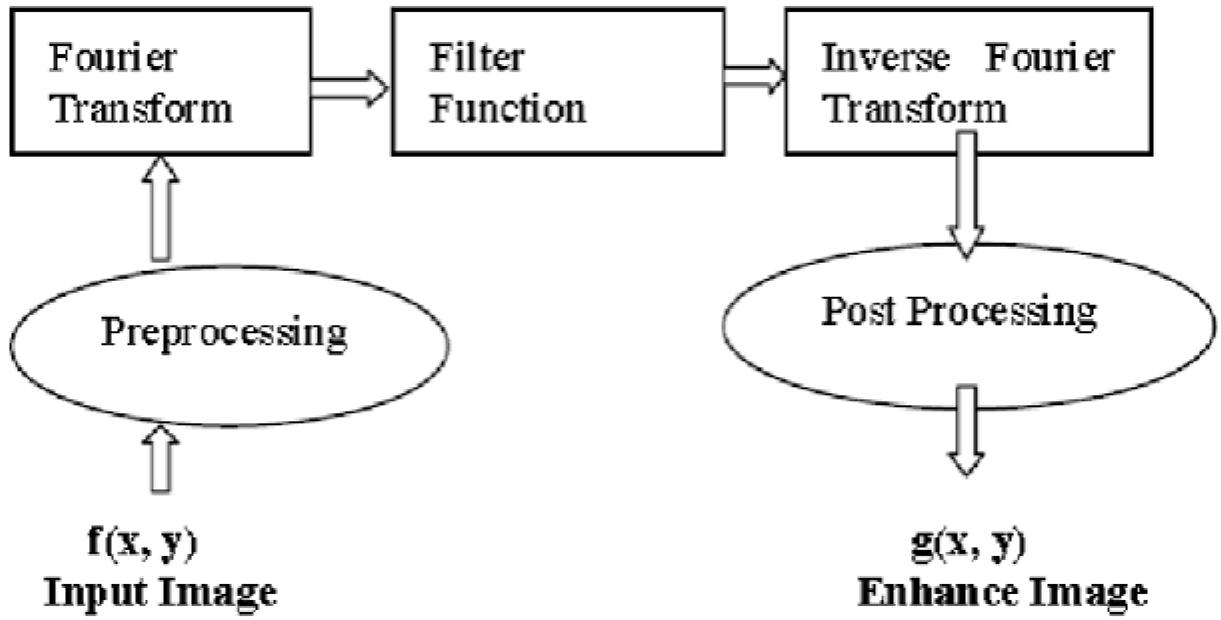
Steps:

- Convert the image into the frequency domain using the **Fourier Transform (FFT)**.
- Locate the periodic noise peaks.
- Remove these peaks using a Notch Filter.
- Apply the **Inverse Fourier Transform** to reconstruct the clean image.



d) Justify why your selected approach is more suitable than spatial-domain methods.

A Notch Filter removes only the unwanted periodic frequency components while preserving the important image details. Spatial-domain filters such as Mean or Median filters remove noise randomly and often blur edges and fine details. Therefore, frequency-domain filtering provides better image quality and higher inspection accuracy.



e) Explain your answer in a logical sequence.

- Capture the metal component image.
- Identify periodic horizontal stripe noise.
- Apply Fourier Transform.
- Detect noise frequencies.
- Remove them using a Notch Filter.
- Apply Inverse Fourier Transform.
- Obtain a clean image.
- Detect defects accurately.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpret the scenario and explain the challenge involved.

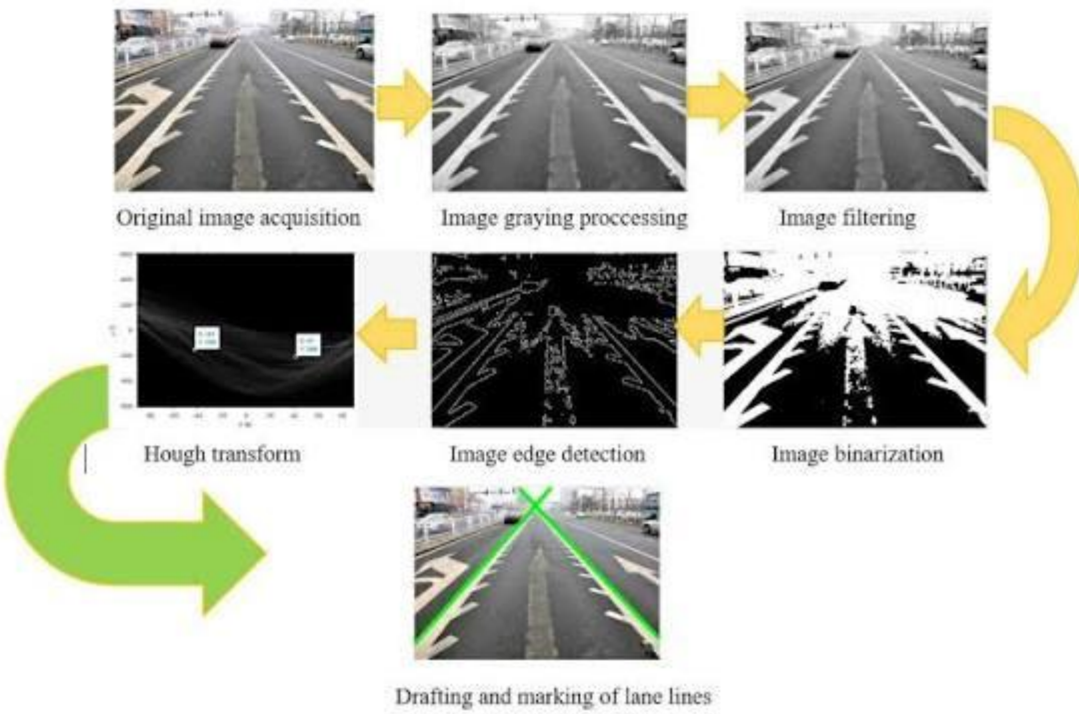
A self-driving car captures road images during heavy rain. Rain introduces random noise into the images while lane markings, road boundaries, and traffic signs must remain clear for safe navigation.

(b) Identify the importance of preserving edges while removing noise.

Edges represent important information such as:

- Lane markings
- Road boundaries
- Vehicles
- Pedestrians
- Traffic signs

If edges become blurred, the autonomous vehicle may fail to recognize lanes or obstacles, increasing the risk of accidents.



(c) Recommend a suitable filtering technique.

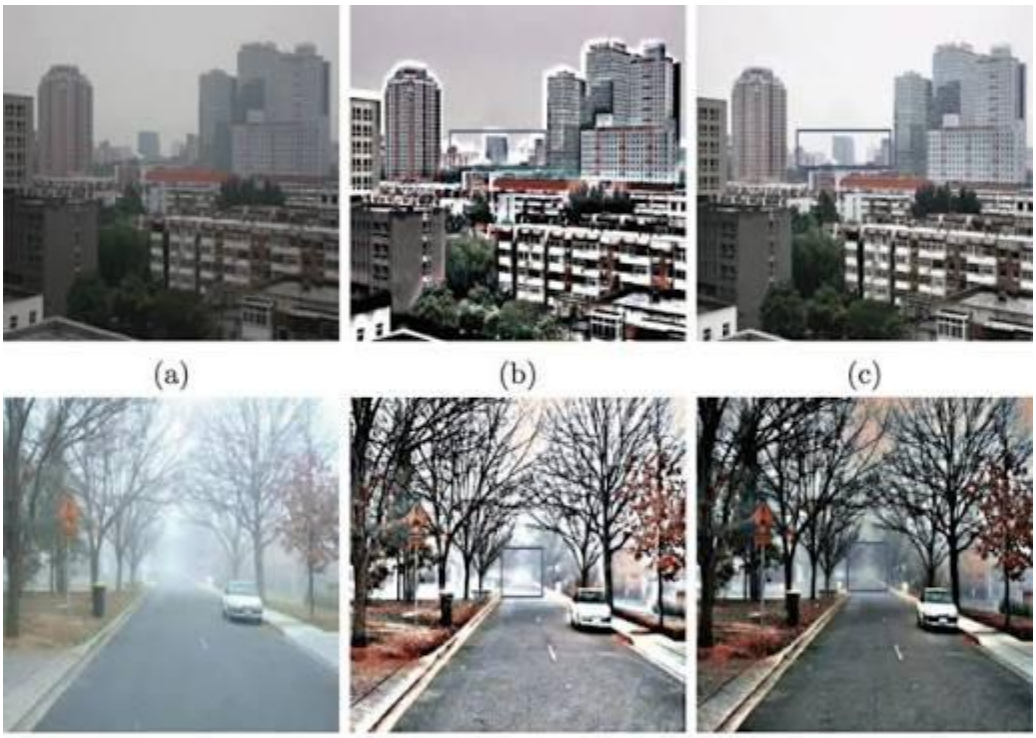
The best filtering technique is the **Bilateral Filter**.

Working:

- Removes noise.
- Preserves edges.
- Considers both spatial distance and pixel intensity differences.
- Produces smooth images without blurring object boundaries.

(d) Justify why your selected filter preserves important image details.

Unlike Gaussian or Mean filtering, the Bilateral Filter smooths only similar neighboring pixels while maintaining sharp intensity changes at object boundaries. Therefore, lane markings and road edges remain clear, improving autonomous driving performance.



(a) (b) (c)

(d) (e) (f)

(e) Present your explanation clearly and systematically.

1. Capture road image during rain.
2. Identify rain noise.
3. Apply Bilateral Filter.
4. Remove unwanted noise.
5. Preserve lane markings and road edges.
6. Detect roads and obstacles accurately.
7. Ensure safe autonomous navigation.